

Autonomous UAV System for Human Detection and Tracking Using YOLOv8 and Real-Time Navigation

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Abstract—This paper presents a lightweight, fully autonomous Unmanned Aerial Vehicle (UAV) system designed to enhance Search and Rescue (SAR) missions through onboard human detection, tracking, and adaptive navigation. Traditional SAR methods are often slow, resource-intensive, and limited by terrain and environmental hazards. In contrast, UAVs offer rapid deployment and wide-area coverage but typically rely on ground control or cloud-based inference, introducing latency and operational constraints. To address these challenges, the proposed system runs a YOLOv8 object detection model and DeepSORT tracking algorithm entirely onboard a Raspberry Pi 5. An onboard camera captures aerial imagery for real-time inference without external computation, and the Raspberry Pi communicates with the UAV's flight controller via MAVLink to dynamically adjust waypoint missions. DroneKit manages autonomous navigation and behavior switching based on detection outcomes. The system was validated through QGroundControl and Software-In-The-Loop (SITL) simulations, as well as partial field tests to assess detection accuracy under real-world conditions. Results demonstrate effective onboard detection, responsive path adjustment, and reliable integration of tracking and control logic. This work highlights the potential of low-cost, AI-enabled UAV platforms to support time-critical SAR operations with increased autonomy and minimal reliance on external infrastructure.

Index Terms—YOLOv8, UAV, Search and Rescue, DeepSORT, DroneKit, Human Detection, Tracking, Waypoint Navigation

I. INTRODUCTION

Unmanned Aerial Vehicles (UAVs), or drones, have emerged as valuable tools in Search and Rescue (SAR) operations due to their ability to quickly deploy, access remote or hazardous areas, and provide real-time aerial data. Compared to traditional SAR methods which rely on ground teams, helicopters, or trained animals UAVs offer faster response times, broader coverage, and reduced risk to human responders [1], [2]. Their integration into SAR missions has significantly enhanced efficiency and operational safety.

Despite these advantages many UAV systems depend on manual operation or cloud-based data processing, which can

introduce delays and reduce autonomy, particularly in areas with limited connectivity [3], [4].

To address these challenges, this paper proposes an autonomous UAV system that integrates a real-time vision-based detection and tracking solution. The system is built using a Raspberry Pi 5 as the onboard processing unit, running a YOLO (You Only Look Once) deep learning model for human detection. The first part of the project focuses on enabling onboard inference and visual recognition using a camera-equipped drone platform. The second part involves designing and implementing a waypoint-based navigation system enhanced with dynamic tracking capabilities using DeepSORT. The complete system is validated through simulation to demonstrate its effectiveness in detecting and tracking humans while autonomously adapting its flight mission.

This work contributes toward improving real-time autonomy in UAV-assisted SAR by eliminating dependence on external computing and enabling decision-making directly on the drone.

II. LITERATURE REVIEW

The evolution of autonomous UAVs for SAR missions has been shaped by advances in artificial intelligence (AI), embedded computing, and real-time vision systems. In SAR contexts—where timely detection of individuals and rapid decision-making are crucial UAVs have transitioned from passive, remote-controlled devices to intelligent aerial agents capable of autonomous operation.

A foundational component of this transformation is real-time object detection, primarily enabled by Convolutional Neural Networks (CNNs). CNNs have demonstrated strong performance in identifying humans from aerial perspectives, even under varying lighting or environmental conditions [5]–[7]. Among detection frameworks, the You Only Look Once (YOLO) family of models has gained popularity due to its balance of speed and accuracy. YOLOv8, the latest iteration,

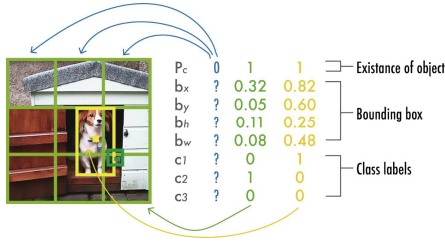


Fig. 1. YOLO model with a three-by-three grid [9].

offers structural improvements and is well-suited for deployment on lightweight embedded systems like the Raspberry Pi 5 [8].

YOLO divides the input image into an $S \times S$ grid, where each grid cell predicts bounding boxes, their confidence scores, and the class probabilities for objects within that cell. The Fig. 1 shows a simplified YOLO model with a three-by-three grid, three classes, and a single class prediction per grid element to produce a vector of eight values [9]. This grid-based design allows YOLO to efficiently process the entire image in a single pass [10]. Despite its impressive speed and accuracy, YOLO's performance is affected by a number of variables, including the training dataset's quality and the surrounding environment. Improving the model with SAR-specific datasets and adapting it to the hardware limitations of UAVs are crucial steps in overcoming these obstacles. For aerial applications, to address this, researchers commonly use custom datasets that better represent the operating environment. In [11]–[13], all three studies created or adapted UAV-specific datasets containing human figures captured from different altitudes, backgrounds, and lighting conditions. These datasets often include augmented images to simulate real-world variation and improve generalization.

Additionally, detection alone is insufficient for dynamic SAR operations. Persistent tracking of detected targets is necessary to avoid loss of visual continuity, especially when both the UAV and the target are in motion. DeepSORT, a widely adopted tracking algorithm, complements YOLO-based detection by assigning consistent identities across frames—allowing the system to follow a specific individual over time.

Another critical factor is the ability to run these computations onboard. While many earlier UAV-based SAR systems offloaded detection tasks to ground stations or cloud services, such approaches suffer from latency and communication dependencies. The use of Single Board Computers (SBCs) like the Raspberry Pi 5 enables edge-based inference, where AI tasks are executed directly on the UAV, reducing response time and increasing operational autonomy.

To navigate effectively during SAR missions, UAVs must integrate visual feedback with flight control. Libraries such as DroneKit allow mission plans to be dynamically modified based on visual input, supporting waypoint-based navigation and mid-mission behavioral changes [14]. Combined with flight controllers like Pixhawk and GPS/IMU modules [15],

these systems can carry out complex patrol or search patterns while responding to live detection events.

Together, these technologies form the foundation for building a UAV that not only identifies humans but also autonomously tracks them and alters its flight path accordingly. The reviewed literature highlights that while object detection and tracking algorithms have matured, there is still limited integration of these components into a fully onboard, lightweight, and adaptable SAR platform. This gap motivates the current work, which combines vision-based detection and tracking with autonomous waypoint navigation in a unified, real-time system validated through simulation.

A. Related Work

Table I summarizes five key works in AI-assisted UAV systems, highlighting their contributions, methodologies, and limitations in relation to the proposed approach.

As shown in Table I, most prior work focused either on detection-only capabilities or required offboard/cloud processing. Very few integrate all key components — detection, tracking, decision-making, and autonomous flight — into a single, lightweight onboard system. The proposed system addresses this gap by using a YOLOv8 model and DeepSORT tracker on a Raspberry Pi 5, allowing dynamic mission behavior using DroneKit.

III. METHODOLOGY

A. Hardware

The UAV system was built using the S500 quadcopter frame equipped with a Pixhawk 2.4.8 flight controller. The platform was later modified to integrate additional components, including a Raspberry Pi 5 for onboard processing and a camera module for real-time image capture. The Raspberry Pi handles image inference and communicates with the Pixhawk, which is responsible for flight stabilization and GPS-based navigation. An overview of the complete hardware setup is shown in Fig. 2.

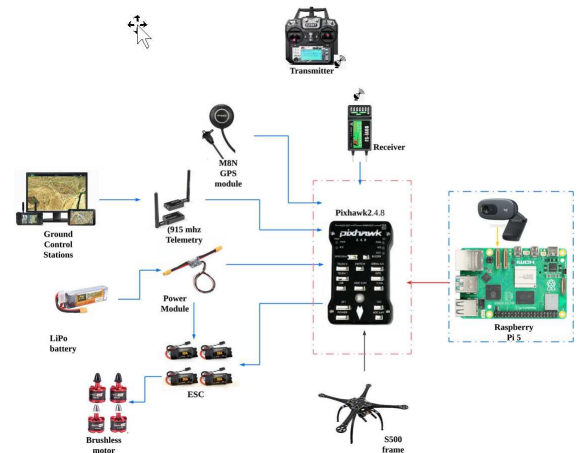


Fig. 2. Overall Drone Hardware Setup.

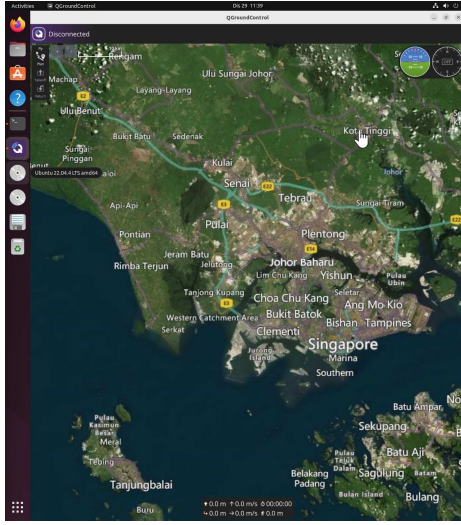


Fig. 5. QGroundControl Interface.

IV. RESULTS AND DISCUSSION

To ensure accurate human identification from aerial views, a custom YOLOv8n model was trained using a relabeled open-source dataset from Roboflow, containing annotated aerial human images. The training process was conducted on Google Colab to leverage GPU acceleration and cloud-based resources. After 40 epochs, the model achieved a mAP@50 of 0.68 and a mAP@50–95 of 0.39, while precision and recall reached 0.72 and 0.61, respectively. The resulting training and validation metrics are shown in Fig. 6, and examples of predicted outputs from the trained YOLOv8n model are presented in Fig. 7.

The trained model was deployed on a Raspberry Pi 5 with DeepSORT for real-time video processing. Remote-controlled drone tests were conducted to validate detection stability. The system consistently detected humans during flight as shown in the below Fig. 8, 9 and 10 confirming its suitability for aerial search and rescue applications.

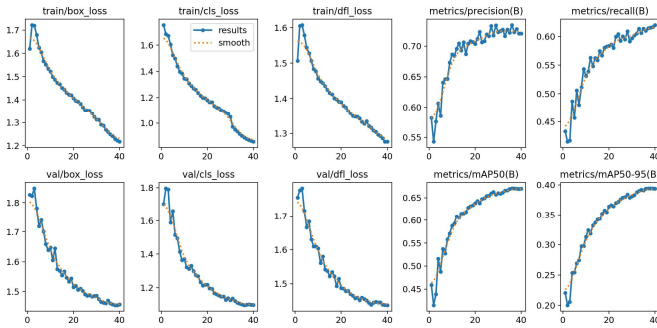


Fig. 6. YOLOv8 training and validation metrics.



Fig. 7. Predicted outputs from the trained YOLOv8n model.



Fig. 8. Detection of a walking human with a confidence score of 0.83.

After confirming successful human detection and tracking, a simple control algorithm was developed to align the detected human with the center of the drone's field of view. This was achieved by adjusting the drone's yaw when the detected human was horizontally off-center, and moving the drone forward when the bounding box size (which correlates with distance) was below a predefined threshold.

The positional offset between the center of the image frame and the center of the annotated bounding box is defined as:

$$\Delta x = x_c - x_f \quad \text{and} \quad \Delta y = y_c - y_f \quad (1)$$

where x_c , y_c represent the coordinates of the bounding box center, and x_f , y_f denote the center of the image frame.

The control logic is defined as:

$$\begin{aligned} \text{If } |\Delta x| > T_x &\Rightarrow \text{Adjust Yaw} \\ \text{If } |\Delta y| > T_y &\Rightarrow \text{Move Forward/Backward} \end{aligned} \quad (2)$$



Fig. 9. Detection near a building edge with partial occlusion.

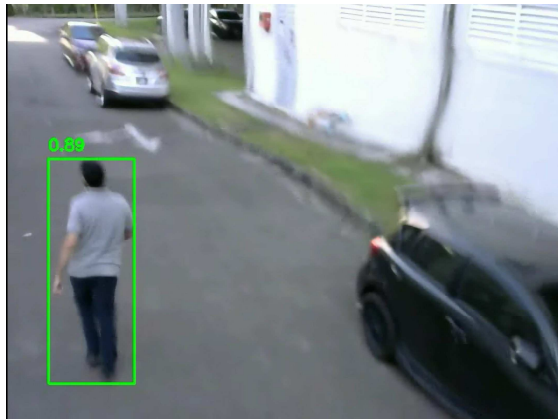


Fig. 10. Rear-view human detection with high confidence under variable lighting conditions.

To validate the behavior of this tracking algorithm during autonomous waypoint missions, simulations were conducted using QGroundControl (QGC) and Software-In-The-Loop (SITL). The system dynamically responds to human detection results by adjusting the drone's movement in real time.

At any point during the tracking process, the user retains the ability to override or terminate tracking through a user interface, with the following options; i. Resume the original waypoint-following mission (skip tracking), ii. Trigger a Return-to-Launch (RTL) sequence to safely return the drone to its takeoff location.

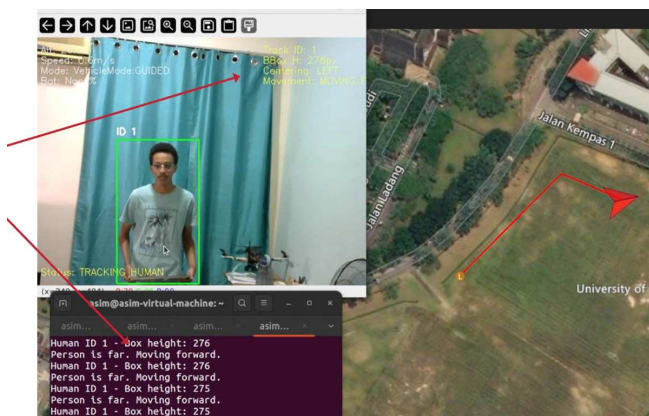


Fig. 11. Drone moves forward as the detected human is far and not centered.

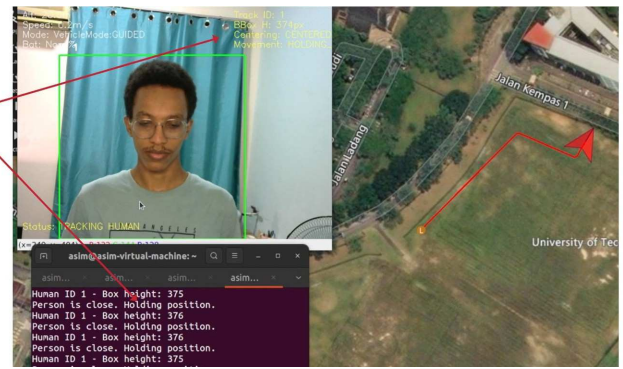


Fig. 12. Drone holds its position as the detected human is near and centered.

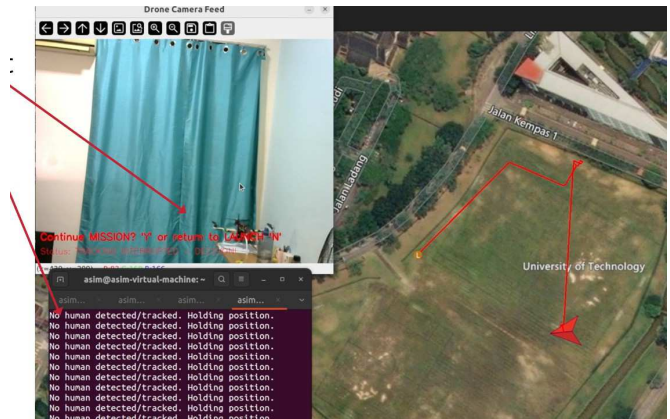


Fig. 13. Drone pauses and holds position as human tracking is lost.

Fig. 11, 12 and 13 shows the human detection respond accordingly. These interactive feature enhances both safety and flexibility, especially in unpredictable or dynamic mission environments. During simulation testing, key tracking metrics such as confidence score, bounding box height, and center offset were analyzed, as shown in Fig. 14. Results indicated that higher confidence values aligned with larger bounding boxes, reflecting closer and more accurate detections, while larger subjects were generally centered in the frame. Most confident detections occurred within moderate offset ranges, confirming effective tracking logic. Overall, the system demonstrated stable and reliable multi-target tracking suitable for autonomous following tasks.

The results presented confirm the effectiveness of the proposed UAV system. The integration of stable flight control, onboard real-time detection, and autonomous human-following behavior forms a complete and reliable SAR solution. The system demonstrates strong performance in simulation environment, making it a promising platform for real world search and rescue applications that demand autonomy, adaptability, and dependable detection under resource-constrained conditions.

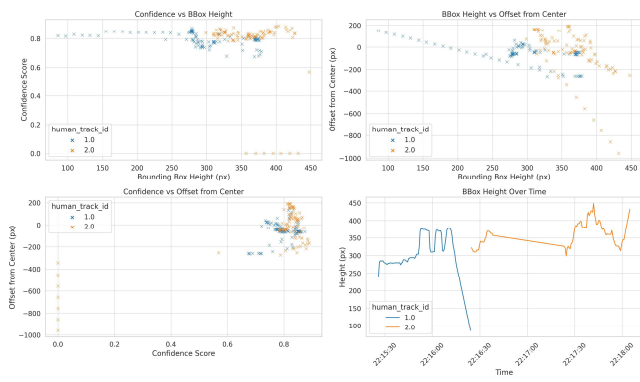


Fig. 14. Correlated tracking metrics across bounding box height, offset from center, confidence score, and time for two tracked human IDs.

V. CONCLUSION

This paper presents a compact UAV system capable of accurate human detection, reliable tracking, and responsive mission control, as achieved through the development of a complete software pipeline. This included training a custom YOLOv8 model for real-time human detection, deploying it on a Raspberry Pi, and validating its performance during actual drone operations. The system demonstrated reliable data exchange with the flight controller via MAVLink, enabling responsive decision-making based on live visual input. Simulation tests confirmed autonomous waypoint navigation using DroneKit and QGroundControl, while a real-time tracking and centering algorithm allowed the drone to dynamically respond to human presence. Furthermore, the system supports user interaction during flight, enabling manual override, return-to-launch (RTL), or continuation of the mission, ensuring both flexibility and safety. For future work, it is recommended to upgrade the vision system to a depth camera to improve detection and environmental awareness, and to adopt a more optimized drone frame for improved stability, efficiency, and overall flight performance.

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